

L10: Deep Learning for Timeseries

CSci 560 Deep Learning w/ Python (Chollet) Ch. 10

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Summer 2025

L10.1 Different Kinds of Timeseries Tasks

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Timeseries Datasets

- A **timeseries** can be any data obtained via measurements at regular intervals.
- Examples include:
 - Daily price of a stock
 - Hourly electricity consumption of a city
 - Weekly sales of a store
 - Speech input sound waves captured at 8kHz (8,000 sample per second)
- Working with timeseries involves understanding the **dynamics** of a system:
 - its periodic cycles
 - how it trends over time
 - its regular regime and its sudden spikes

Kinds of Timeseries Tasks

- **Forecasting:** predicting what will happen next in a timeseries.
 - The most common, for example forecast electricity consumption a few hours in advance so can anticipate demand.
- **Classification:** Assign one or more categorical labels to a timeseries.
 - Given timeseries activity of a visitor on a website, classify whether visitor is bot or human
- **Event detection:** Identify the occurrence of a specific expected event within a continuous data stream.
 - *hotword detection* to monitor a stream and detect utterances like “Hey Alexa”
- **Anomaly detection:** Detect anything unusual happening within a continuous datastream.
 - Unusual credit card timeseries may be fraud

L10.2 A Temperature Forecasting Example

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L10.3 Understanding Recurrent Neural Networks

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Understanding Recurrent Neural Networks

- All of the neural networks you've seen so far have no memory.
 - Each input shown is processed independently with no state kept between inputs.
 - For a dense network, we flatten five days of data into a single large vector: **feedforward networks**.
- Biological systems use memory
 - while reading you process each word one-by-one while keeping memories of what came before.
- A **recurrent neural network** (RNN) adopts the same principle.
 - Processes sequences by iterating through the sequence elements and maintaining a **state**

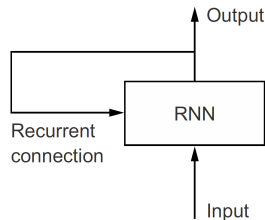


Figure 1: A recurrent network: a network with a loop. (Chollet, 2021, pg. 294)

Recurrent Operation

- A **recurrent neural network** (RNN) adopts the same principle.
 - Processes sequences by iterating through the sequence elements and maintaining a **state**
- The state of the RNN is reset between processing two different sequences, so you still consider one sequence to be a single input.
- What changes is that this data point is no longer processed in a single step
 - rather, the network internally **loops** over sequence elements

Example implementation of forward pass of a toy RNN:

```
# the state at time t
state_t = 0

# iterate over sequence elements
for input_t in input_sequence:
    output_t = f(input_t, state_t)
    # the previous output becomes the state for the next iteration
    state_t = output_t
```



Recurrent Operation

To be more specific, you can even flesh out the function f . It is a tensor operation on the `input_t` and the `state_t`.

```
# the state at time t
state_t = 0

# iterate over sequence elements
for input_t in input_sequence:
    output_t = activation(dot(W, input_t) + dot(U, state_t) + b)
    # the previous output becomes the state for the next iteration
    state_t = output_t
```



RNN Step Function

A RNN is a for loop that reuses quantites computed during the previous iteration of the loop.

RNNs are characterized by their step function:

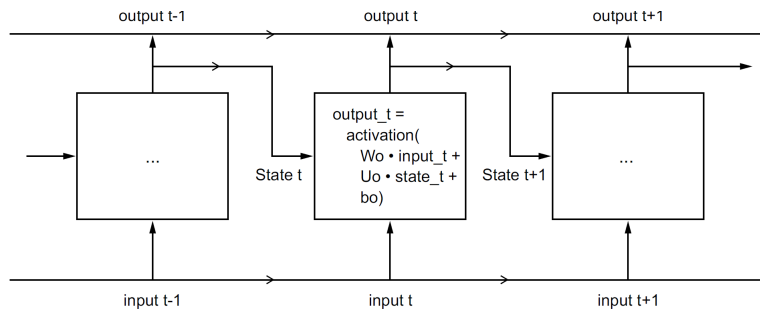


Figure 2: A simple RNN, unrolled over time. (Chollet, 2021, pg.295)

RNN Layer Output in Keras

- All recurrent layers in Keras SimpleRNN, LSTM and GRU can be run in two different modes
 - ① they can return the full sequences of successive outputs for each time step like we did before, a rank-3 tensor of shape `(batch_size, timesteps, output_features)`
 - ② or they can return only the last output for each input sequence a rank-2 tensor of shape `(batch_size, output_features)`
- These two modes are controlled by the `return_sequences` constructor arguments.

Limits of the SimpleRNN Layer

- In practice, you'll rarely work with the SimpleRNN layer.
- SimpleRNN has a major issue
 - **vanishing gradient** problem
 - same as we discussed for deep networks
 - each loop in a SimpleRNN acts like a layer, so if a large number of loops, gradients diminish and are hard to get signal through a simple RNN when performing weight update.
- LSTM and GRU essentially have inner components similar to residual connections to mitigate vanishing gradients.

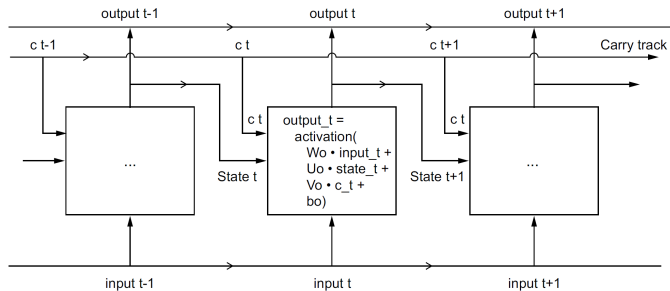


LSTM Layer: Avoiding Vanishing Gradients in Recurrent Layers

- A variant of the 'SimpleRNN
- But it adds a way to carry information across many timesteps.
- Essentially what LSTM does is to save information for later thus preventing older signals from gradually vanishing during processing.
 - e.g. similar to residual connections

Going from SimpleRNN to the LSTM

- W_o , U_o and b_o are the weights at $output_t$
- LSTM adds in additional dataflow c_t where C stands for **carry**.
- The carry has the following impact
 - it will be combined with the input connection and the recurrent connection followed by a bias add and the application of an activation function
 - and it will affect the state being sent to the next timestep
- Conceptually the carry dataflow is a way to modulate the next output and the next state.



Going from SimpleRNN to the LSTM: output and i,f,k activations

- The carry dataflow is computed using 3 distinct transformations.
- All three have form of a SimpleRNN cell transformation.
- We'll index with letters i f and k

```
output_t = activation(dot(state_t, Uo) + dot(input_t, Wo) + dot(c_t, Vo) + bo)
i_t = activation(dot(state_t, Ui) + dot(input_t, Wi) + bi)
f_t = activation(dot(state_t, Uf) + dot(input_t, Wf) + bf)
k_t = activation(dot(state_t, Uk) + dot(input_t, Wi) + bk)
```

Going from SimpleRNN to the LSTM: compute the carry

We obtain the new carry state (the next c_t) by combining i_t , f_t and k_t

$$c_{t+1} = c_t * i_t * k_t * f_t$$

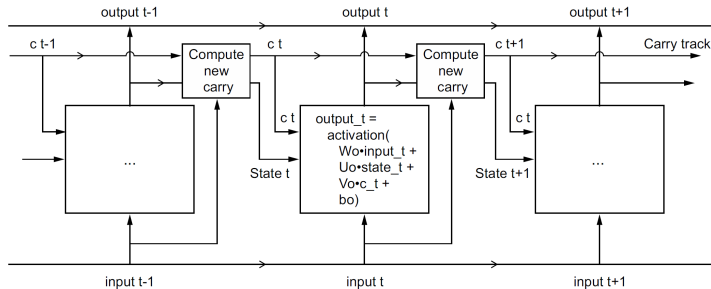


Figure 4: Anatomy of an LSTM. (Chollet, 2021, pg.299)

What these operations **actually** do is determined by the contents of the weights parameterizing them; and the weights are learned in an end-to-end fashion, starting over with each training round.

L10.4 Advanced Use of Recurrent Neural Networks

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Bidirectional RNN

- A bidirectional RNN exploits the order sensitivity of RNNs
- It uses two regular RNNs
 - One sees input in normal chronological order
 - The other sees it in reverse chronological order
- A bidirectional RNN exploits the idea that, for some problems, reverse order is a different but useful presentation to build representations.
- Bidirectional RNNs are a great fit for text data, or any other kind of data where order matters, yet which order you use doesn't matter.

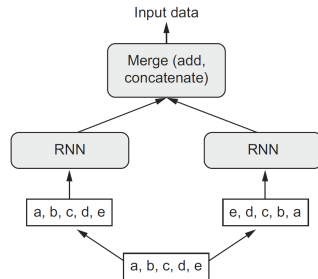


Figure 5: How a bidirectional RNN layer works. (Chollet, 2021, pg.306)

Summary of Deep Learning for Timeseries

- It's good to first establish common-sense baselines for your metric of choice. If you don't have a baseline to beat, you can't tell whether you're making real progress.
- Try simple models before expensive ones, to make sure the additional expense is justified. Sometimes a simple model will turn out to be your best option.
- When you have data where ordering matters, and in particular for timeseries data, recurrent networks are a great fit and easily outperform models that first flatten the temporal data.
- The two essential RNN layers available in Keras are the LSTM layer and the GRU layer.
- To use dropout with recurrent networks, you should use a time-constant dropout mask and recurrent dropout mask.
 - These are built into Keras recurrent layers, so all you have to do is use the `recurrent_dropout` arguments.
- Stacked RNNs provide more representational power than a single RNN layer. They're also much more expensive and thus not always worth it.

Chollet, F. (2021). *Deep learning with python* (second). [Manning](#).